SI 140A-01 Probability & Statistics for EECS, Fall 2026
Homework 3

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Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

To battle against spam, Bob installs two anti-spam programs. An email arrives, which is either legitimate (event L) or spam (event L^c), and which program j marks as legitimate (event M_j) or marks as spam (event M_j^c) for $j \in \{1, 2\}$. Assume that 10% of Bobs email is legitimate and that the two programs are each "90% accurate" in the sense that $P(M_j|L) = P(M_j^c|L^c) = 9/10$. Also assume that given whether an email is spam, the two programs' outputs are conditionally independent.

(a) Find the probability that the email is legitimate, given that the 1st program marks it as legitimate (simplify).

(b) Find the probability that the email is legitimate, given that both programs mark it as legitimate (simplify).

(c) Bob runs the 1st program and M_1 occurs. He updates his probabilities and then runs the 2nd program. Let $\tilde{P}(A) = P(A|M_1)$ be the updated probability function after running the 1st program. Explain briefly in words whether or not $\tilde{P}(L|M_2) = P(L|M_1 \cap M_2)$: is conditioning on $M_1 \cap M_2$ in one step equivalent to first conditioning on M_1, then updating probabilities, and then conditioning on M_2?

Solution

(a) By Bayes' rule,

$$\begin{aligned} P(L \mid M_1) &= \frac{P(M_1 \mid L)P(L)}{P(M_1)} \\ &= \frac{P(M_1 \mid L)P(L)}{P(M_1 \mid L)P(L) + P(M_1 \mid L^c)P(L^c)} \\ &= \frac{0.9 \times 0.1}{0.9 \times 0.1 + 0.1 \times 0.9} = \frac{0.09}{0.18} = \frac{1}{2}. \end{aligned}$$

(b) We have

$$P(L \mid M_1 \cap M_2) = \frac{P(M_1 \cap M_2 \mid L)P(L)}{P(M_1 \cap M_2)}.$$

By conditional independence,

$$P(M_1 \cap M_2 \mid L) = 0.9 \times 0.9 = 0.81.$$

Also,

$$\begin{aligned} P(M_1 \cap M_2) &= P(M_1 \cap M_2 \mid L)P(L) + P(M_1 \cap M_2 \mid L^c)P(L^c) \\ &= 0.81 \times 0.1 + 0.1 \times 0.1 \times 0.9 = 0.09. \end{aligned}$$

Thus,

$$P(L \mid M_1 \cap M_2) = \frac{0.81 \times 0.1}{0.09} = 0.9 = \frac{9}{10}.$$

Problem 1 continued on next page…

(c) We have

$$\begin{aligned}\widetilde{P}(L \mid M_2) &= \frac{\widetilde{P}(L \cap M_2)}{\widetilde{P}(M_2)} \\ &= \frac{P(L \cap M_2 \mid M_1)}{P(M_2 \mid M_1)} \\ &= \frac{P(L \cap M_2 \cap M_1)/P(M_1)}{P(M_1 \cap M_2)/P(M_1)} \\ &= P(L \mid M_1 \cap M_2).\end{aligned}$$

So they are equal. Since $\widetilde{P}$ already incorporates the information from M_1, conditioning further on M_2 is equivalent to conditioning on $M_1 \cap M_2$ in one step.

Problem 2

Fred decides to take a series of n tests, to diagnose whether he has a certain disease (any individual test is not perfectly reliable, so he hopes to reduce his uncertainty by taking multiple tests). Let D be the event that he has the disease, $p = P(D)$ be the prior probability that he has the disease, and $q = 1 - p$. Let T_j be the event that he tests positive on the jth test.

(a) Assume for this part that the test results are conditionally independent given Fred's disease status. Let $a = P(T_j \mid D)$ and $b = P(T_j \mid D^c)$, where a and b don't depend on the jth test. Find the posterior probability that Fred has the disease, given that he tests positive on all n of the n tests.

(b) Suppose that Fred tests positive on all n tests. However, some people have a certain gene that makes them always test positive. Let G be the event that Fred has the gene. Assume that $P(G) = 1/2$ and that D and G are independent. If Fred does not have the gene, then the test results are conditionally independent given his disease status. Let $a_0 = P(T_j \mid D, G^c)$ and $b_0 = P(T_j \mid D^c, G^c)$, where a_0 and b_0 don't depend on j. Find the posterior probability that Fred has the disease, given that he tests positive on all n of the tests.

Solution

(a) Define

$$E = \bigcap_{j=1}^{n} T_j,$$

the event that Fred tests positive on all n tests. By Bayes' rule,

$$P(D \mid E) = \frac{P(E \mid D)P(D)}{P(E \mid D)P(D) + P(E \mid D^c)P(D^c)}.$$

By conditional independence,

$$P(E \mid D) = \prod_{j=1}^{n} P(T_j \mid D) = a^n, \qquad P(E \mid D^c) = b^n.$$

Therefore,

$$P(D \mid E) = \frac{pa^n}{pa^n + qb^n}.$$

(b) Since $P(G) = \frac{1}{2}$ and D and G are independent,

$$P(D \cap G) = P(D \cap G^c) = \frac{p}{2}, \qquad P(D^c \cap G) = P(D^c \cap G^c) = \frac{q}{2}.$$

We also have

$$P(E \mid D, G) = P(E \mid D^c, G) = 1,$$

and

$$P(E \mid D, G^c) = a_0^n, \qquad P(E \mid D^c, G^c) = b_0^n.$$

Problem 2 continued on next page…

Divide the population according to gene status. Then

$$\begin{aligned} P(D \cap E) &= P(D \cap G)P(E \mid D, G) + P(D \cap G^c)P(E \mid D, G^c) \\ &= \frac{p}{2} + \frac{p}{2}a_0^n = \frac{p}{2}(1 + a_0^n). \end{aligned}$$

Similarly,

$$P(D^c \cap E) = \frac{q}{2}(1 + b_0^n), \qquad P(E) = \frac{p}{2}(1 + a_0^n) + \frac{q}{2}(1 + b_0^n).$$

Therefore,

$$\begin{aligned} P(D \mid E) &= \frac{P(D \cap E)}{P(E)} \\ &= \frac{p(1 + a_0^n)}{p(1 + a_0^n) + q(1 + b_0^n)} \\ &= \frac{p(1 + a_0^n)}{1 + pa_0^n + qb_0^n}. \end{aligned}$$

Problem 3

Suppose that in the population of gamers, being good at Genshin Impact is independent of being good at Apex (with respect to some measure of "good"). A certain school club has a simple admission procedure: admit an applicant if and only if the applicant is good at Genshin Impact or is good at Apex.

(a) Intuitive explain why it makes sense that among gamers that the club admits, being good at Apex is negatively associated with being good at Genshin Impact, *i.e.*, conditioning on being good at Apex decreases the chance of being good at Genshin Impact.

(b) (Berkson's paradox) Show that if A and B are independent and $C = A \cup B$, then A and B are conditionally dependent given C (as long as $P(A \cap B) > 0$ and $P(A \cup B) < 1$), with

$$P(A \mid B \cap C) < P(A \mid C).$$

Solution

(a) Among admitted gamers, one must be good at at least one of the games. Since being good at Genshin Impact and being good at Apex are independent, if a gamer is good at Genshin Impact, there is already a sufficient explanation for admission. Therefore, the evidence that the gamer is good at Apex becomes weaker.

(b) We have $C = A \cup B$, A and B are independent, and

$$A = C \cap A, \qquad B = C \cap B.$$

Thus,

$$P(A \mid B \cap C) = P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A).$$

Also,

$$P(A \mid C) = \frac{P(A \cap C)}{P(C)} = \frac{P(A)}{P(C)}.$$

Since $P(A) > 0$ and $0 < P(C) < 1$,

$$\frac{P(A)}{P(C)} > P(A).$$

Therefore,

$$P(A \mid B \cap C) < P(A \mid C).$$

Problem 4

You are going to play 2 games of chess with an opponent whom you have never played against before (for the sake of this problem). Your opponent is equally likely to be a beginner, intermediate, or a master. Depending on which, your chances of winning an individual game are 90%, 50%, or 30%, respectively.

(a) What is your probability of winning the first game?

(b) Congratulations: you won the first game! Given this information, what is the probability that you will also win the second game (assume that, given the skill level of your opponent, the outcomes of the games are independent)?

(c) Explain the distinction between assuming that the outcomes of the games are independent and assuming that they are conditionally independent given the opponent' s skill level. Which of these assumptions seems more reasonable, and why?

Solution

(a) Let B, I, and M be the events that the opponent is a beginner, intermediate, or a master, respectively. Let W_j be the event of winning the jth game. Then

$$P(B) = P(I) = P(M) = \frac{1}{3},$$

and

$$P(W_j \mid B) = 0.9, \qquad P(W_j \mid I) = 0.5, \qquad P(W_j \mid M) = 0.3.$$

Hence,

$$\begin{aligned} P(W_1) &= P(W_1 \mid B)P(B) + P(W_1 \mid I)P(I) + P(W_1 \mid M)P(M) \\ &= \frac{1}{3}(0.9 + 0.5 + 0.3) = \frac{17}{30} \approx 56.67\%. \end{aligned}$$

(b) By Bayes' rule,

$$P(B \mid W_1) = \frac{P(W_1 \mid B)P(B)}{P(W_1)} = \frac{0.9 \times \frac{1}{3}}{17/30} = \frac{9}{17}.$$

Similarly,

$$P(I \mid W_1) = \frac{5}{17}, \qquad P(M \mid W_1) = \frac{3}{17}.$$

Given the opponent's skill level, the outcomes are conditionally independent:

$$P(W_2 \mid W_1 \cap H) = P(W_2 \mid H), \qquad H \in \{B, I, M\}.$$

Thus,

$$\begin{aligned} P(W_2 \mid W_1) &= \sum_{H \in \{B,I,M\}} P(W_2 \mid H)P(H \mid W_1) \\ &= \frac{9}{17} \times 0.9 + \frac{5}{17} \times 0.5 + \frac{3}{17} \times 0.3 \\ &= \frac{11.5}{17} = \frac{23}{34} \approx 67.65\%. \end{aligned}$$

Problem 4 continued on next page…

(c) Unconditional independence means

$$P(W_2 \mid W_1) = P(W_2).$$

Winning the first game would not change the probability of winning the second game.

Conditional independence means

$$P(W_2 \mid W_1 \cap H) = P(W_2 \mid H), \qquad H \in \{B, I, M\}.$$

If the skill level is initially unknown, the first game's outcome can provide information and change the probability.

Conditional independence is more reasonable because we play against the same opponent, whose unknown skill level affects the outcomes. Winning the first game implies that the opponent is more likely to be a beginner, thereby increasing the estimated probability of winning the second game.